

Global plasma modeling of an air-breathing gridded ion thruster for very low Earth orbit applications

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A volume-averaged (zero-dimensional) global model of an atmospheric nitrogen-oxygen mixture is proposed. This model is used to predict the performance of an air-breathing electric gridded-ion thruster. Based on particle and energy balance equations, it computes the densities and temperatures of the different species in the discharge chamber and evaluates the thrust generated by ions extracted from the plasma and accelerated between two grids. Simulation results over a range of input radio-frequency (RF) powers and altitudes are used to assess the feasible operating regimes and to identify the main limiting mechanisms. Coupling the global model with a simplified drag analysis indicates that satellite operation should be feasible at altitudes between 200 and 250 km, for RF power levels ranging from 1.5 to 3 kW. However, certain limitations associated with the gas compression system and drag estimation deserve further investigation.

I. INTRODUCTION

Operating satellites in very low Earth orbit (VLEO), i. e. 150-400 km, promises sharper imaging, lower latency, reduced radiation exposure, and natural end-of-life disposal, but it also brings persistent aerodynamic drag that conventional (stored-propellant) systems cannot offset indefinitely. Air-breathing electric propulsion (ABEP) addresses this constraint by collecting the residual upper-atmosphere flow with an intake and using an electric thruster to ionize and accelerate that gas to produce drag-compensating thrust.¹ In practice, the ABEP concept is a tightly coupled system: intake collection and transmission set the available mass flow, which is used to feed the discharge. Plasma production and acceleration set the thrust at a given power provided by solar arrays. Since the external environment varies along the orbit, it complicates control and power management.

On the intake side, recent studies consider both specular (free-molecular) and diffuse (passive) scattering models²⁻⁵. In these studies, the predicted transmission and compression vary widely depending on the intake geometry and how surface interactions are modeled⁶. A recent design⁷ integrating both an intake and a compatible air-fed thruster illustrates the central trade-off: maximizing useful mass capture without adding prohibitive drag or complexity. On the thruster side, ABEP architectures have explored Radio-frequency (RF) gridded ion devices⁸, Hall-type variants^{9,10}, air-breathing electron cyclotron resonance (ECR) ion engines^{11,12}, and inductive plasma thrusters (IPTs)^{7,13} or microwave sources¹⁴. Early demonstrations with atmospheric propellants showed that ion sources can operate on nitrogen-oxygen mixtures¹, but performance margins are tight in VLEO: system analyses indicate that, for a representative small platform, maintaining long-term orbital stability near 190-200 km requires thrust-to-power ratios on the order of 8 mN kW^{-1} or higher and specific impulses in the multi-kilosecond range¹⁵. Alongside these performance constraints, technology gaps persist: most notably, air-fed neutralization and material compatibility with atomic oxygen¹.

In recent years, several studies have developed models to analyze drag compensation in ABEP scenarios. In particular, the work of Giannetti *et al.*¹⁶ investigates drag compensation from a system-level perspective. By evaluating the efficiencies of the different subsystems, the authors identify key parameters governing ABEP feasibility as a function of altitude, platform geometry, and thruster performance. In a similar fashion, other studies¹⁷ analyze the thrust-to-drag balance by determining the vehicle parameters (e.g., spacecraft frontal area, propellant exhaust velocity)

required for sustained operation in VLEO. These works establish the minimum operational altitude as a function of spacecraft and thruster characteristics.

A reliable way to explore the thruster design-space at early stages is to use volume-averaged (“0D”/global) plasma models that balance reaction and energy budgets to predict discharge states for given geometry, gas composition, and power. Such models are widely used in electric propulsion and plasma physics to capture first-order scaling without resolving full spatial detail¹⁸. They have been formulated and validated for RF-driven gridded ion thrusters (GITs) using well-characterized propellants: most notably xenon¹⁹ and iodine^{20–22}, demonstrating that carefully constructed reaction sets and electron-energy treatments can reproduce measured plasma parameters and thrust trends. Similar efforts on air have been done by Mrózek *et al.*²³ on a magnetized RF plasma with atmospheric mixtures and by Lopez-Uricoechea *et al.*⁸ on N_2 – O_2 mixtures in a radiofrequency ion thruster.

In this work, we develop a zero-dimensional (global) model of atmospheric air, including multi-ion chemistry, to support the sizing of an ABEP plasma chamber and to evaluate ABEP feasibility under VLEO conditions. The model describes an RF discharge in air by explicitly tracking the main neutral and ionic species and by solving coupled particle and energy balance equations. This allows us to predict the electron temperature, ionization fraction, and species densities as functions of chamber geometry (effective radius and length), pressure, which is set by the intake mass flow, and input RF power. The model is conceived as a first-order tool to assess the performance of a thruster that employs biased grids to accelerate ions extracted from an inductively coupled plasma (ICP) discharge. In this configuration, the atmospheric air collected from the external environment is injected into the chamber, where it is ionized by an oscillating electromagnetic field at the standard radio frequency of 13.56 MHz²⁴. The approach adopted in this work is similar to that of Lopez-Uricoechea *et al.*⁸, with several notable differences: the mass flow rate is solely determined by the particles collected at a given altitude and cannot be independently adjusted; the altitude range considered here lies above 150 km, whereas their study focuses on lower altitudes; and the reaction dataset employed differs.

To validate the approach, we first apply the model to a xenon discharge and benchmark its predictions against the well-established results of Chabert *et al.*¹⁹. Subsequently, we validate our model for pure-nitrogen and pure-oxygen discharges, comparing our results to the experimental measurements of Singh & Graves²⁵, and with the global model developed by Thorsteinsson & Gudmunsonn¹⁸. Building on this reference case, the model is then extended to atmospheric air,

whose composition is determined by studying the VLEO atmosphere, by incorporating a comprehensive set of chemical reactions involving nitrogen, oxygen, and their ionic derivatives. In this way, the model captures the chemical complexity of air plasmas and solves, at each time step, for the density and temperature of the dominant species.

From these states, we estimate wall and volume losses, ion flux to extraction grids, and the extractable ion current. These quantities are directly connected to thrust and thrust-to-power when coupled with simple grid-extraction relations. This approach follows the established methodology of global modeling (reaction/energy closure with geometry-dependent losses) but adapts it to the mixture and charge-state complexity of air, which differs qualitatively from noble-gas or iodine cases¹⁸. The result is a fast tool to map ignition margins, attainable ion currents, and power partitioning over altitude-dependent mass-flow envelopes; it highlights where ABEP operation is plausible and where limitations will likely dominate. While such a 0D model cannot predict spatial non-uniformities, detailed RF coupling, or plume evolution, it is well-suited for first-order sizing and for narrowing the large ABEP design space before higher-fidelity (DSMC for intakes, PIC for thrusters) and hardware testing are committed¹⁹.

This paper is organized as follows: in Section II, we present the atmospheric conditions of the environment. The 0D plasma discharge model is described and validated in Section III. In Section IV, we use our model to examine the possibility of using an air-fed gridded ion thruster to balance the atmospheric drag. We expose our final remarks in Section V.

II. ATMOSPHERIC CONDITIONS AT VLEO

The composition of the atmosphere varies significantly with altitude. In order to calculate the inputs of the model, one needs to estimate the densities of molecules and atoms at different altitudes. The composition of the atmosphere, reported in Figure 1, at the orbit altitude is determined using the NRLMSISE-00 model²⁶ developed by NASA. In this model, the Earth's magnetic field, solar radiation, and the composition of the atmosphere are considered to be strictly altitude-dependent.

Figure 1 (a) shows the atmospheric species densities as a function of altitude, computed on 1 January 2025 (00:00 UTC) along a circular orbit at constant altitude, with latitude and longitude varying according to the orbital motion (inclination 0°), assuming moderate solar activity ($F_{10.7} = F_{10.7A} = 150$ sfu) and quiet geomagnetic conditions ($A_p = 4$). As one can see, between 150 and

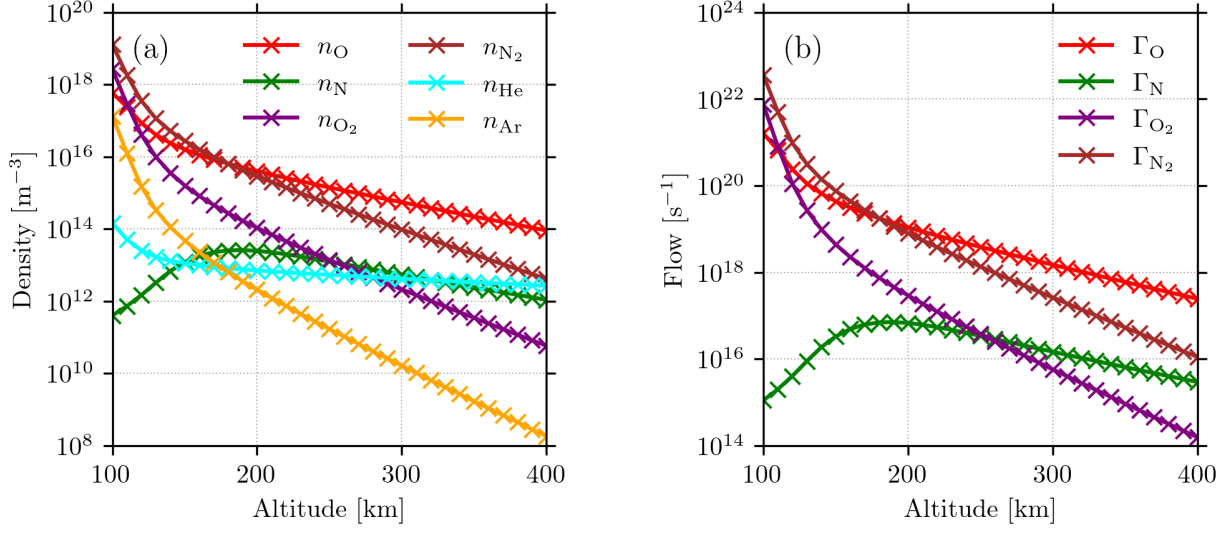


FIG. 1: (a) Evolution of the densities of the species found in VLEO atmosphere, with respect to altitude, calculated with the NRLMSISE-00 model²⁶. (b) Collected flows by a 1 m^2 intake with collection efficiency of 35%.

300 km, the gas is mostly formed by atomic and molecular nitrogen and oxygen, while the other components are minor. Hence, in the following, we neglect argon and helium. While the present results provide an initial assessment of the feasibility of ABEP drag compensation, a complete evaluation would require incorporating the effects of solar-activity-driven variability. Addressing this factor lies beyond the scope of the current work.

To estimate the particles' encounter rate with the spacecraft, we multiply the orbital speed by the ambient number density and the effective cross-sectional area of the spacecraft (i.e., 1 m^2 here), yielding the injection rate in units of particles per second. The orbital velocity at a given altitude was computed assuming a circular orbit, using the Earth's standard gravitational parameter ($\mu = 3.986 \times 10^{14} \text{ m}^3 \text{ s}^{-2}$) and an Earth radius of $R_{\text{Earth}} = 6371 \text{ km}$. The orbital speed was calculated as

$$v = \sqrt{\frac{\mu}{R_{\text{Earth}} + a}}, \quad (1)$$

where a denotes the orbital altitude. The results for the four main components are reported in Figure 1 (b) for a collection efficiency η_c of 35%¹. We assume that a fraction η_c of the incoming particle flux is effectively injected into the discharge chamber. To make our results comparable with those of standard, already operational thrusters, we further assume that there is no backflow through the back wall inlet of the chamber. This condition is difficult to achieve using passive

compression alone, and may therefore require active gas compression²⁷. However, in this study we focus exclusively on the discharge region. A detailed analysis of particle inlet and backflow processes is left for future work.

III. ZERO-DIMENSIONAL MODEL

We develop a zero-dimensional global model of the plasma, describing its time-dependent thermochemical behavior via energy and particle balance equations for all major atomic, molecular, ionic, and electronic species. The model includes both bulk physico-chemical reactions (ionization, excitation, dissociation, etc.) and surface-related losses (wall losses and recombination, particle flux through the grid at one end, etc.).

A distinctive feature of this model is the extension of classical single-ion global models for gridded ion thrusters¹⁹ to handle a realistic multi-ion composition typical of atmospheric air plasmas. The model assumes a non-equilibrium thermal distribution: electrons, monatomic species, and diatomic species each follow separate Maxwellian distributions with distinct temperatures T_e , T_{mono} , and T_{diat} , respectively. In this work, we assume thermal equilibrium among all monatomic species, and separately among all diatomic species. This choice follows the approach of²¹ and²², who consider two distinct equilibria based on particle masses.

A. Chemical species and reactions

The plasma considered in this work includes electrons (e^-), molecular nitrogen (N_2), molecular oxygen (O_2), atomic nitrogen (N), atomic oxygen (O), and their primary singly-charged ions (N_2^+ , O_2^+ , N^+ , O^+). In line with the discussion in Section II, this choice reflects literature consensus indicating that these species dominate at typical electron temperatures (i. e. T_e of a few eV) and low-pressure (i. e. below 1 mTorr) conditions found in ABEP scenarios²⁸. Moreover, we neglect multiply charged ions, as previously done in RF GIT global modeling¹⁹. We also omit negatively charged ions: low-pressure pure nitrogen discharges (without considering O^-/O_2^-) are electropositive and well captured in global models^{18,29}. Furthermore, at pressure roughly equal to 0.5 mTorr, with an electron temperature of a few eV, the net attachment pathways in air are supposed to be weak. Therefore, it is reasonable to leave the effect of electronegativity for a future study. In the same way, termolecular reactions scale with density³⁰ and are negligible in our pressure regime.

B. Model Principles

For a species X , two balance equations are solved: a particle balance,

$$\frac{dn_X}{dt} = \sum_j \delta n_{X,j} \mathcal{R}_j^X,$$

and an energy balance,

$$\frac{d}{dt} (c_{v,X} n_X e T_X) = \sum_j \delta \mathcal{E}_{X,j} \mathcal{R}_j^X,$$

where n_X and T_X denote the density and temperature of species X , respectively, $\mathcal{R}_j^X$ is the rate of reaction j , and $\delta n_{X,j}$ and $\delta \mathcal{E}_{X,j}$ are the corresponding particles (e.g. $\delta n_{X,j} = \pm 1$) and energy variations. The heat capacity $c_{v,X}$ is equal to $3/2$ for monatomic particles, and $7/2$ for diatomic molecules to account for additional rotational and vibrational degrees of freedom. The particle balance is solved for all species (i. e. electrons, all types of ions, and all types of neutrals), while the energy balance is solved only for electrons, monatomic and diatomic species, as proposed in Ref.²²

The electron energy balance, which is central due to RF heating, reads

$$\frac{d}{dt} \left(\frac{3}{2} n_e e T_e \right) = P_{\text{abs}} - P_{\text{loss}},$$

where P_{abs} is the volumetric power absorbed via inductive coupling, and P_{loss} includes inelastic and elastic collision losses, wall losses, and recombination processes. The considered processes and reactions are detailed in Section III E.

Assuming all reactions are elementary, the reaction rate of a collision j of an electron with a neutral of species X is modeled as:

$$\mathcal{R}_j^X = K_j(T_e) n_e n_X, \quad (2)$$

where n_e is the electron density, n_X the density of the neutral species, and $K_j(T_e)$ is the rate constant depending on the electron temperature T_e . Each type of reaction has an associated cross section $\sigma_j(E)$. The rate constant is computed by averaging over a distribution of electron velocities, as

$$K_j(T_e) = \int_{v(E_{\text{th}})}^{+\infty} \sigma_j(v(E)) f(v(E), T_e) v(E) dv, \quad (3)$$

where E_{th} is the threshold energy of the reaction, $v(E) = \sqrt{2E/m_e}$, and $f(v(E), T_e)$ is a normalized Maxwellian distribution, in this case. The quantity m_e represents the electron mass. Cross-section data are taken from databases referenced in Table I and Table II, inspired by Zhou's work²⁸.

TABLE I: Nitrogen reactions.

Label	Collision	ϵ_{th} (eV)	Reference
ion01_N2	$e + \text{N}_2(X^1\Sigma_g^+) \rightarrow 2e + \text{N}_2^+$	15.60	31,32
exc1_N2	$e + \text{N}_2(X^1\Sigma_g^+) \rightarrow e + \text{N}_2(A^3\Sigma_u^+)$	6.17	31,32
exc2_N2	$e + \text{N}_2(X^1\Sigma_g^+) \rightarrow e + \text{N}_2(B^3\Pi_g)$	7.35	31,32
exc3_N2	$e + \text{N}_2(X^1\Sigma_g^+) \rightarrow e + \text{N}_2(W^3\Delta_u)$	7.36	31,32
exc4_N2	$e + \text{N}_2(X^1\Sigma_g^+) \rightarrow e + \text{N}_2(B'^3\Sigma_u^-)$	8.16	31,32
exc5_N2	$e + \text{N}_2(X^1\Sigma_g^+) \rightarrow e + \text{N}_2(a'^1\Sigma_u^-)$	8.40	31,32
exc6_N2	$e + \text{N}_2(X^1\Sigma_g^+) \rightarrow e + \text{N}_2(a^1\Pi_g)$	8.55	31,32
exc7_N2	$e + \text{N}_2(X^1\Sigma_g^+) \rightarrow e + \text{N}_2(w^1\Delta_u)$	8.89	31,32
exc8_N2	$e + \text{N}_2(X^1\Sigma_g^+) \rightarrow e + \text{N}_2(b^1\Pi_u)$	12.50	31,32
exc9_N2	$e + \text{N}_2(X^1\Sigma_g^+) \rightarrow e + \text{N}_2(b'^1\Sigma_u^+)$	12.90	31,32
exc11_N2	$e + \text{N}_2(X^1\Sigma_g^+) \rightarrow e + \text{N}_2(c'^1\Sigma_u^+)$	12.90	31,32
exc12_N2	$e + \text{N}_2(X^1\Sigma_g^+) \rightarrow e + \text{N}_2(C^3\Pi_u)$	11.00	31,32
exc13_N2	$e + \text{N}_2(X^1\Sigma_g^+) \rightarrow e + \text{N}_2(E^3\Sigma_g^+)$	11.90	31,32
exc14_N2	$e + \text{N}_2(X^1\Sigma_g^+) \rightarrow e + \text{N}_2(a''^1\Sigma_g^+)$	12.30	31,32
ela_N2	$e + \text{N}_2(X^1\Sigma_g^+) \rightarrow e + \text{N}_2(X^1\Sigma_g^+)$	0	31,32
diss_N2	$e + \text{N}_2(X^1\Sigma_g^+) \rightarrow e + \text{N} + \text{N}$	12.30	31,32
rot_N2	$e + \text{N}_2(X^1\Sigma_g^+, J \geq 0) \rightarrow e + \text{N}_2(X^1\Sigma_g^+, J = \text{All})$	0	33
vib_N2	$e + \text{N}_2(X^1\Sigma_g^+, v = 0) \rightarrow e + \text{N}_2(X^1\Sigma_g^+, v = \text{All})$	0.01	31,32
ion01_N	$e + \text{N}(^4\text{S}) \rightarrow 2e + \text{N}^+$	14.54	34
exc1_N	$e + \text{N}(^4\text{S}) \rightarrow e + \text{N}(^2\text{D})$	2.39	34
exc2_N	$e + \text{N}(^4\text{S}) \rightarrow e + \text{N}(^2\text{P})$	3.58	34
ela_N	$e + \text{N}(^4\text{S}) \rightarrow e + \text{N}(^4\text{S})$	0	34

C. Chamber geometry and boundary conditions

The plasma chamber is modeled as a cylinder characterized by its radius R and length L , bounded axially by a wall at one end and by extraction grids at the other. The total volume is V . A schematic of the discharge chamber is presented in Figure 2. This design, similar to that of Chabert *et al.*¹⁹, does not include a particle backflow towards the intake.

TABLE II: Oxygen reactions.

Label	Collision	ϵ_{th} (eV)	Reference
ion01_O2	$e + \text{O}_2(X^3\Sigma_g^-) \rightarrow 2e + \text{O}_2^+$	12.10	³⁵
exc1_O2	$e + \text{O}_2(X^3\Sigma_g^-) \rightarrow e + \text{O}_2(a^1\Delta_g)$	0.98	³⁵
exc2_O2	$e + \text{O}_2(X^3\Sigma_g^-) \rightarrow e + \text{O}_2(b^1\Sigma_g^+)$	1.63	³⁵
exc3_O2	$e + \text{O}_2(X^3\Sigma_g^-) \rightarrow e + \text{O}_2(A^3\Sigma_u^+)$	4.50	³⁶
exc4_O2	$e + \text{O}_2(X^3\Sigma_g^-) \rightarrow e + \text{O}_2(B^3\Sigma_u^-)$	6.10	³⁶
ela_O2	$e + \text{O}_2(X^3\Sigma_g^-) \rightarrow e + \text{O}_2(X^3\Sigma_g^-)$	0	³⁷
diss1_O2	$e + \text{O}_2(X^3\Sigma_g^-) \rightarrow e + \text{O}(^3\text{P}) + \text{O}(^3\text{P})$	6.00	³⁸
diss2_O2	$e + \text{O}_2(X^3\Sigma_g^-) \rightarrow e + \text{O}(^3\text{P}) + \text{O}(^1\text{D})$	8.40	³⁸
rot_O2	$e + \text{O}_2(X^3\Sigma_g^-, J \geq 0) \rightarrow e + \text{O}_2(X^3\Sigma_g^-, J = \text{All})$	0	³³
vib_O2	$e + \text{O}_2(X^3\Sigma_g^-, v = 0) \rightarrow e + \text{O}_2(X^3\Sigma_g^-, v = \text{All})$	0.20	³⁵
ion01_O	$e + \text{O}(^3\text{P}) \rightarrow 2e + \text{O}^+$	13.60	³⁵
exc1_O	$e + \text{O}(^3\text{P}) \rightarrow e + \text{O}(^1\text{D})$	1.97	³⁵
exc2_O	$e + \text{O}(^3\text{P}) \rightarrow e + \text{O}(^1\text{S})$	4.19	³⁵
exc3_O	$e + \text{O}(^3\text{P}) \rightarrow e + \text{O}(3\text{S}^0)$	9.52	³⁹
exc4-9_O	$e + \text{O}(^3\text{P}) \rightarrow e + \text{O}^*$	> 12	³⁹
ela_O	$e + \text{O}(^3\text{P}) \rightarrow e + \text{O}(^3\text{P})$	0	³⁵

We apply classical sheath theory for modeling plasma-wall interactions: ions reach the chamber walls and grid apertures at the Bohm speed, $u_{B,X} = \sqrt{eT_e/m_X}$, with m_X the mass of the species X and e the elementary charge. This condition satisfies the criterion for a multi-species electropositive plasma introduced by Riemann⁴⁰ and is discussed below. Floating-wall boundary conditions are applied. Equilibrium between electron and ion fluxes towards surfaces^{24,41} forms at the walls. Due to the confinement in a chamber, density gradients develop within the plasma, requiring geometrical correction factors h_R (radial) and h_L (axial) to account for edge-to-center plasma density ratios. Following the formulation of Ref.¹⁹, we express these factors as a function of the ion-neutral mean-free paths λ ⁴², as:

$$h_R = 0.8 \left(4 + \frac{R}{\lambda}\right)^{-1/2}, \quad h_L = 0.86 \left(3 + \frac{L}{2\lambda}\right)^{-1/2}.$$

In the previous expression, we consider a generic $\sigma_i = 1 \times 10^{-18} \text{ m}^2$ ^{43,44}, as a global cross section

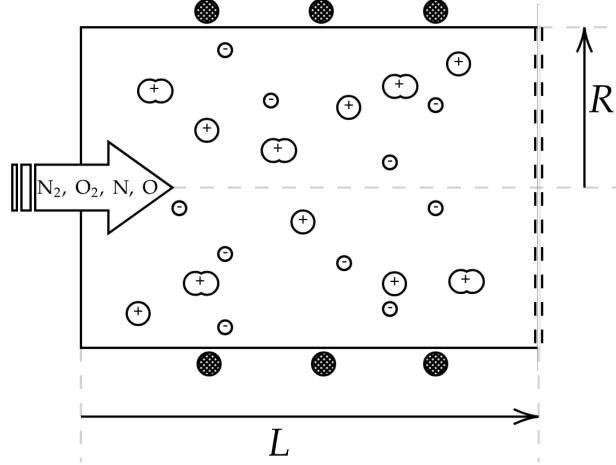


FIG. 2: Schematic of the plasma thruster model. An RF coil operating at 13.56 MHz surrounds the cylindrical chamber. Neutral gas is injected from the left, ions are accelerated by biased grids, and un-ionized propellant exits through the grid holes.

including charge exchange and elastic scattering collisions for every species, to calculate the mean-free-path as $\lambda = (\sigma_i n_{g,tot})^{-1}$. The parameter $n_{g,tot}$ is the total neutral density. Even if this choice allows us to obtain the correct order of magnitude of the mean free path, one should notice that more refined models can be used¹⁸. However, this goes beyond the scope of the present work.

The plasma extraction from the discharge chamber, which produces the thrust, is obtained using a biased grid. The partial transparency of the grid to ions and neutrals is modeled by the parameters β_{ion} and β_{neu} , respectively, quantifying the fraction of particles passing through the grid without reflection or recombination. Accordingly, the effective surface areas relevant to wall interactions and particle losses in the balance equations are defined as

$$\begin{aligned} S_{eff} &= h_R L 2\pi R + 2h_L \pi R^2, \\ S_{extr} &= h_L \pi R^2, \\ S_{open,neu} &= \beta_{neu} \pi R^2, \\ S_{open,ions} &= \beta_{ion} h_L \pi R^2, \\ S_{neutr,ions} &= h_R L 2\pi R + (2 - \beta_{ion}) h_L \pi R^2. \end{aligned}$$

Here, S_{eff} represents the effective wall-loss surface area for ions and electrons, S_{extr} is the area of the cylinder base, $S_{open,neu}$ and $S_{open,ions}$ denote the effective open areas of the extraction grid for neutrals and ions, respectively, and $S_{neutr,ions}$ corresponds to the surface area available for ion

recombination into neutral atoms and molecules at the chamber walls. In the present work, we do not analyze the coupling of the biased grid with the plume neutralizer, that, however, might have an impact on the potential near the grid.

D. Thrust calculation

The thrust generated by the ion species X^+ is calculated as

$$F_{T,X^+} = \Gamma_{X^+} m_{X^+} v_{\text{beam},X} S_{\text{open,ions}}, \quad (4)$$

with

$$\Gamma_{X^+} = n_{X^+} u_{B,X^+},$$

$$v_{\text{beam},X} = \sqrt{\frac{2eV_{\text{acc}}}{m_{X^+}}}.$$

V_{acc} is the voltage bias between the positive screen grid and the neutralizer. In real devices, the voltage drop between internal and external grid is expected to be $\sim 20\%$ larger than V_{acc} .

The thrust of the neutral species X is

$$F_{T,X} = \Gamma_X m_X v_X S_{\text{open,neu}}. \quad (5)$$

The neutral velocity is

$$v_X = \sqrt{\frac{8k_B T_X}{\pi m_X}},$$

and the flux

$$\Gamma_X = \frac{1}{4} n_X v_X.$$

The total thrust $F_{T,\text{tot}}$ is obtained as the sum of neutral and ion thrust of all species. Beam divergence and ions impinging on the acceleration grid are neglected. We can also define the total thrust power generated by ions and neutrals, P_I and P_N , respectively, as²⁴

$$P_I = \sum_X e \Gamma_{X^+} S_{\text{open,ions}} V_{\text{acc}}, \quad (6)$$

$$P_N = \sum_X \frac{1}{2} m_X v_X^2 \Gamma_X S_{\text{open,neu}}. \quad (7)$$

E. Plasma global model details

We present here the complete set of volumetric and surface reactions considered in our global plasma model, explicitly detailing their contributions to particle and energy balances, following the approaches of Chabert *et al.*¹⁹ and Esteves *et al.*²².

1. Volumetric Reactions

Neutral gas injection is modeled as a volumetric source. The density n_X of the neutral species X reads

$$\left. \frac{dn_X}{dt} \right|_{\text{inj}} = \frac{Q_X}{V}.$$

The particle injection rate Q_X , depending on the altitude, is imposed and constant in time for every neutral species X , as described in Section II. The injection temperature T_0 is equal for all species. The neutral species X energy increases by

$$\left. \frac{d}{dt} (c_{v,X} n_X e T_X) \right|_{\text{inj}} = c_{v,X} \frac{Q_X}{V} e T_0.$$

We recall that the quantity T_X indicates the temperature of the species X expressed in eV.

As discussed in Section III B, we model separately the power absorbed and lost by the electrons. When a current I_{coil} flows, the RF power absorbed by the plasma reads $P_{\text{abs}} = \frac{1}{2} R_{\text{ind}} I_{\text{coil}}^2$, with inductive impedance

$$R_{\text{ind}} = \frac{2\pi N^2}{L\omega\epsilon_0} \Re \left(\frac{\iota k R J_1(k_p R)}{\epsilon_p J_0(k_p R)} \right)$$

computed as in Ref.²⁴. N is the number of coils' turns, ϵ_0 the vacuum permittivity, and ω the RF frequency. J_0 and J_1 are Bessel functions, with $k = k_0 \epsilon_p^{1/2}$, and $k_0 = \omega/c$. c is the speed of light and ι is the imaginary unit. The complex plasma permittivity ϵ_p incorporates electron-neutral collision frequency ν_m and electron plasma frequency $\omega_{p,e}$, as

$$\epsilon_p = 1 - \frac{\omega_{p,e}^2}{\omega(\omega - \iota \nu_m)}. \quad (8)$$

The total RF power is $P_{\text{RF}} = \frac{1}{2} (R_{\text{ind}} + R_{\text{coil}}) I_{\text{coil}}^2$, where R_{coil} is the resistance of the wire.

Electron-neutral excitation ($e^- + X \rightarrow e^- + X^*$) leads to radiative energy losses as

$$\left. \frac{d}{dt} \left(\frac{3}{2} n_e e T_e \right) \right|_{\text{Exc}} = -E_{\text{exc},X} K_{\text{exc},X} n_e n_X,$$

Atmoprop'X

but no change in densities. $E_{\text{exc},X}$ is the energy lost in the excitation and $K_{\text{exc},X}$ the corresponding rate. Rotation and vibration losses are treated in the same way. Electron-impact ionization ($e^- + X \rightarrow 2e^- + X^+$), with a rate $K_{\text{ion},X}$, increases electron density, and correspondingly the density of the species X^+ , as

$$\left. \frac{dn_e}{dt} \right|_{\text{Ion}} = \left. \frac{dn_{X^+}}{dt} \right|_{\text{Ion}} = n_e n_X K_{\text{ion},X}$$

and reduces the neutral X species density and electron energy, as

$$\left. \frac{dn_X}{dt} \right|_{\text{Ion}} = -n_e n_X K_{\text{ion},X}$$

and

$$\left. \frac{d}{dt} \left(\frac{3}{2} n_e e T_e \right) \right|_{\text{Ion}} = -E_{\text{ion},X} K_{\text{ion},X} n_e n_X,$$

respectively. $E_{\text{ion},X}$ is the energy lost in the ionization process. Elastic electron-neutral collisions redistribute energy between electrons and heavy species with reaction rate $K_{\text{el},X}$, according to

$$\left. \frac{d}{dt} \left(\frac{3}{2} n_e e T_e \right) \right|_{\text{el}} = -\frac{3m_e}{m_X} e (T_e - T_X) K_{\text{el},X} n_e n_X$$

and symmetrically for neutral species X .

Molecular dissociation ($e^- + X_2 \rightarrow e^- + 2X$) alters densities with rate K_{diss} according to

$$\left. \frac{dn_{X_2}}{dt} \right|_{\text{diss}} = -K_{\text{diss}} n_e n_{X_2}$$

and

$$\left. \frac{dn_X}{dt} \right|_{\text{diss}} = 2K_{\text{diss}} n_e n_{X_2},$$

with a corresponding electron energy loss

$$\left. \frac{d}{dt} \left(\frac{3}{2} n_e e T_e \right) \right|_{\text{diss}} = -E_{\text{diss}} K_{\text{diss}} n_e n_{X_2}.$$

The term E_{diss} is the energy cost of the dissociation, i.e. the dissociation threshold. The elastic collisions among heavy species, for each pair of ion Y^+ and neutral X , are described by¹⁹

$$\left. \frac{d}{dt} (c_{v,X} n_X e T_X) \right|_{\text{el,heavy}} = m_{Y^+} u_{B,Y^+}^2 n_{Y^+} n_X \sigma_i \frac{1}{4} \left(\frac{8eT_X}{\pi m_X} \right)^{1/2}. \quad (9)$$

We recall that in the previous expression σ_i is approximated as a constant.

2. Surface Reactions

Neutral species, ions, and electrons are modeled using classical sheath theory²⁴. The flux of neutral species X , which are assumed to follow a Maxwellian distribution, towards the walls is given by

$$\Gamma_{n,X} = \frac{1}{4} n_{g,X} \sqrt{\frac{8eT_{g,X}}{\pi m_X}}. \quad (10)$$

The multi-species Bohm criterion reads⁴⁰

$$\sum_X \frac{e_{X+}^2 n_{X+,0}}{m_X^2 v_{X+,0}^2} \leq \frac{en_{e,0}}{eT_e^2} \quad (11)$$

with e_{X+} the charge of the ion species X^+ and $v_{X+,0}$ the velocity at the sheath edge (indicated by the subscript 0). The densities are also taken at the sheath-edge. The charge neutrality at the sheath-edge also holds. By assuming that each ion species reaches the sheath edge with its respective Bohm velocity, $v_{X+,0} = u_{B,X}$, as previously adopted in Refs.^{20,21}, the inequality in Eq. (11) is satisfied. It should be emphasized, however, that this constitutes only one possible solution, and alternative estimations of the ion velocities at the sheath edge may also be considered. Then, the flux to the walls of the ion of species X^+ with density n_{X+} is given by

$$\Gamma_{i,X+} = n_{X+} u_{B,X}. \quad (12)$$

The edge-to-center ratio, h , depends on the wall type as discussed in Section III C and will be considered within the effective surfaces below. The electron flux to the walls is calculated as the sum of the ionic fluxes on every surface to maintain quasi-neutrality, as

$$\Gamma_e = \sum_{X+} \Gamma_{i,X+}. \quad (13)$$

In order to evaluate the energy losses to the walls in the electron power balance, we need to estimate the potential drop at the walls ϕ_{sheath} . The electron flux to the walls can be written as

$$\Gamma_e = \frac{n_e}{4} \sqrt{\frac{8eT_e}{\pi m_e}} \exp\left(-\frac{\phi_{\text{sheath}}}{T_e}\right).$$

As before, we do not consider the h -factor, which will be included when dealing with the surfaces. By equalizing ion and electron fluxes at steady state, the sheath potential for multi-ion plasmas is obtained as:

$$\phi_{\text{sheath}}^{\beta_w} = T_e \ln \left(\frac{n_e(1 - \beta_w)}{\sqrt{2\pi} \sum_{X+} n_{X+} \sqrt{\frac{m_e}{m_{X+}}}} \right),$$

generalizing the classical sheath potential. This equation depends on the wall type through the parameter β_w . It is equal to β_{ion} on the grid wall and to zero elsewhere. On the grid, the electron flux compensates both the ion flux on the grid and the ion beam flux. This sheath potential determines electron energy losses at the walls, i.e., the mean energy of an electron to pass the potential barrier and reach the wall. It reads

$$E_{\text{loss,e}}^{\beta_w} = e \left(2T_e + \phi_{\text{sheath}}^{\beta_w} \right),$$

when we consider the energy lost by the electrons in the sheath and in the pre-sheath.

Particles lost at walls' boundaries or grids decrease densities and energies, as

$$\frac{dn_e}{dt} \Big|_{\text{wall}} = -\Gamma_e \frac{S_{\text{eff}}}{V},$$

and similarly for neutrals and ions, with the fluxes defined in Eqs. (13) and (10), respectively. The electrons lose additional energy as

$$\frac{d}{dt} \left(\frac{3}{2} n_e e T_e \right) \Big|_{\text{wall}} = -E_{\text{loss,e}}^{\beta_w=0} \Gamma_e \frac{S_{\text{eff}} - S_{\text{extr}}}{V} - E_{\text{loss,e}}^{\beta_w=\beta_{\text{ion}}} \Gamma_e \frac{S_{\text{extr}}}{V}. \quad (14)$$

The energy losses to the walls of the heavy species are treated as thermal losses. These losses are modeled as

$$\frac{d}{dt} (c_{v,X} n_X e T_X) \Big|_{\text{therm,wall}} = -\kappa_X \frac{e}{k_B} \frac{T_X - T_{\text{wall}}}{\Lambda_0} \frac{S_{\text{eff}}}{V}, \quad (15)$$

with Λ_0 the heat diffusion length, such that $\Lambda_0 = \left(\left(\frac{2.405}{R} \right)^2 + \left(\frac{\pi}{L} \right)^2 \right)^{-1/2}$ as in Ref.²¹, and κ_X the thermal conductivity. The wall temperature T_{wall} is taken equal to T_0 . The thermal conductivities of the considered atmospheric species were fitted as power functions of the temperature using Ref.⁴⁵. The thermal coefficients were adjusted to fit the experimental data from Ref.⁴⁶. The following conductivities were obtained and used in this work,

$$\begin{aligned} \kappa_{\text{N}} &= 6.91 \times 10^{-4} \times \left(\frac{e}{k_B} T_{\text{mono}} \right)^{0.6909}, \\ \kappa_{\text{N}_2} &= 3.61 \times 10^{-4} \times \left(\frac{e}{k_B} T_{\text{diat}} \right)^{0.7540}, \\ \kappa_{\text{O}} &= 7.72 \times 10^{-4} \times \left(\frac{e}{k_B} T_{\text{mono}} \right)^{0.6793}, \\ \kappa_{\text{O}_2} &= 2.91 \times 10^{-4} \times \left(\frac{e}{k_B} T_{\text{diat}} \right)^{0.7979}. \end{aligned}$$

In the previous expressions, the temperatures are in eV. The conductivity unit here is W/m/K.

Eventually, the recombination of ions at the wall ($e^- + X^+ \rightarrow X$) replenishes neutrals by

$$\left. \frac{dn_X}{dt} \right|_{\text{rec}} = \Gamma_{X^+} \frac{S_{\text{neutr,ions}}}{V}.$$

We do not consider molecular formation on the chamber walls, as it strongly depends on the wall material and its effect is supposed to be negligible. We use an energy recombination factor of 1: the energy of ions is fully transferred to the surfaces. The secondary electron emission from the walls is neglected.

F. Model validation

To validate our code, several steps were taken. Initially, the code was designed to replicate the results from Chabert *et al.*¹⁹ for xenon. This goal has been achieved and the results in Ref.¹⁹ perfectly reproduced.

In order to test the more complex implementation of molecular dissociation, the validity of chemical reactions and cross sections, etc., we decided to challenge our code with some published results about pure nitrogen and oxygen discharges, for which several references are present. Among them, we decided to use the experimental measurements from Singh & Graves²⁵. In the nitrogen case, these results were also compared to a global model by Thorsteinsson & Gudmunsonn¹⁸. The measurements for nitrogen are taken at constant absorbed RF power of 320 W, in a stainless steel chamber with the dimensions $R = 10\text{cm}$ and $L = 10\text{cm}$. Those for oxygen are taken in the same chamber at constant pressure of 1 mTorr and variable RF power. For nitrogen and oxygen, we use the set of cross-sections in Table I and II, respectively.

By fixing the pressure inside the chamber, we obtained the results for electron temperature and density reported in Figure 3. As one can see, our results successfully reproduce the experimental trend for both gases. It is not surprising that a difference appears in nitrogen with the results of Thorsteinsson & Gudmunsonn¹⁸. The reaction model and cross sections that are used in the current work, discussed in Section III, are not the same that were used by these authors.

As one can see in Figure 3 (a), our code correctly captures the dependency of the electron density on the pressure. In (b), the significant drop in electron temperature when the pressure increases is also captured. One should notice that the electron temperature reaches values almost as high as 10 eV when the pressure falls below the 1 mTorr limit. The results in oxygen also appear accurate: both temperature and density trends and values are well reproduced.

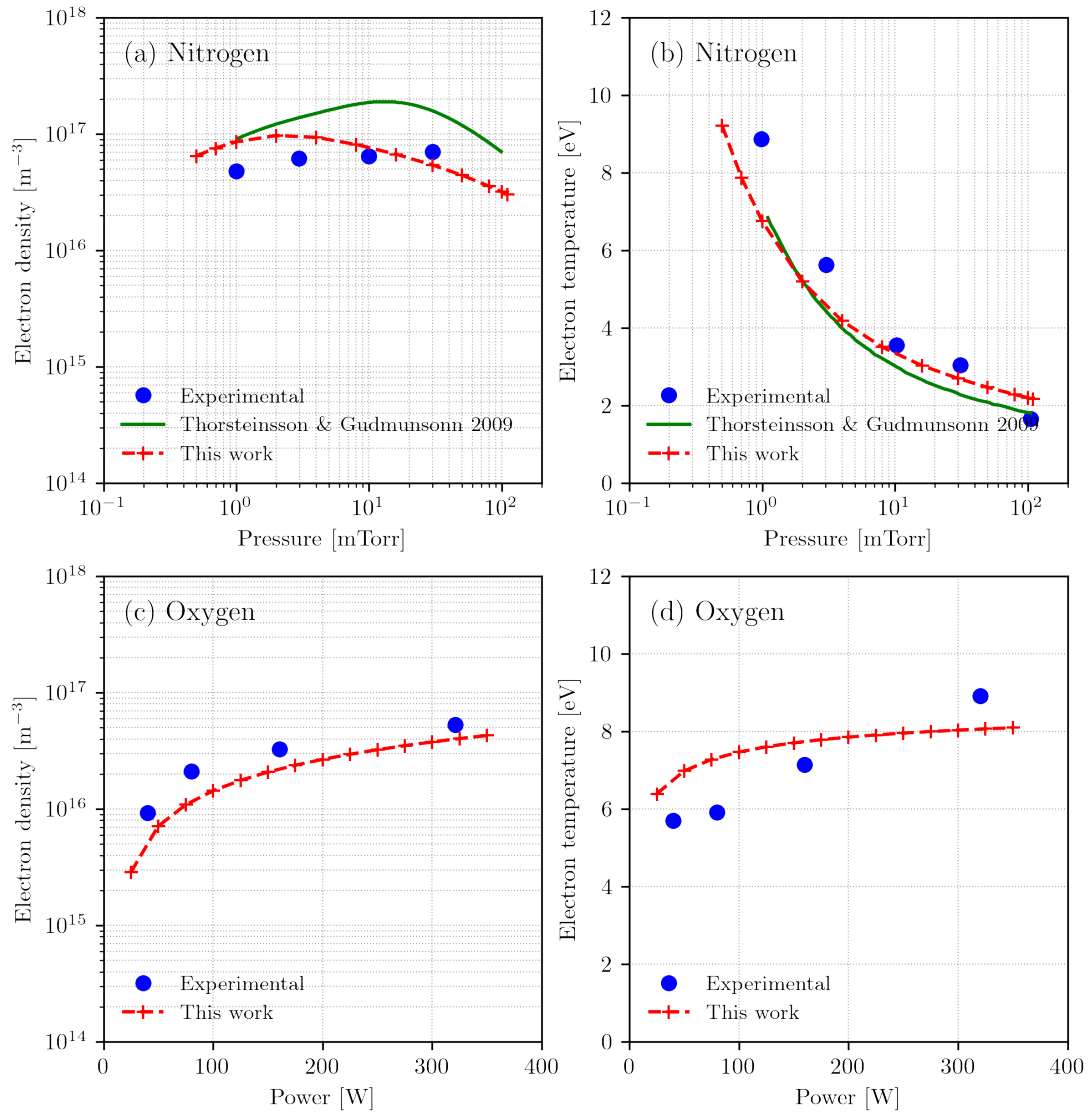


FIG. 3: Comparison of the outcomes of the code developed in the context of the present work, with the experimental results from Singh and Graves²⁵ and the code outcomes of Thorsteinsson & Gudmunsonn¹⁸, when available. A 320 W pure-nitrogen discharge plasma density (a) and temperature (b). A 1 mTorr pure-oxygen discharge density (c) and temperature (d).

IV. RESULTS

In this section, we first analyze the thruster behavior at a fixed altitude of 220 km for a real air mixture for different RF power levels. Later, we study the performances of the thruster as a function of the altitude, again for different RF power levels. The input mass-flow rate is discussed in Section II. We recall that only a fraction of the incoming flux is injected into the discharge chamber and no backflow through the inlet is considered.

A. Temporal transient

Although the model is not intended to describe the transient phase from plasma ignition to steady state, as the initial conditions and ignition process are not modeled in detail, it is still informative to examine the time evolution of the different species in the simulations. The results presented here consider the parameters in Table III at 220 km. To facilitate early-time convergence, we assumed an increased density at $t = 0$ by a factor of 4000 with respect to the atmospheric density. This corresponds to an accumulation of air inside the chamber prior to the plasma ignition. The initial relative abundance of the different species depends only on the altitude. Similarly, the injection rate depends only on the altitude and on the collection area A_{intake} . The heavy species are assumed to be at the same temperature as the external walls at the beginning of the simulation, $T_0 = 350 \text{ K} = 0.03 \text{ eV}$. Clearly, the current model does not allow for evaluating in detail the ignition mechanism, which requires further investigation.

Figure 4 shows the time evolution of the temperature (a) and of the gas and plasma species densities (b). The fixed input RF power is 1500 W and the selected altitude is 220 km. In this figure, we observe that the steady state is reached in slightly more than 1 ms. The variations, both in temperature and densities, happen on the tens of μs scale. In the last hundreds of ms, the solution does not change.

In Figure 4 (a), we observe that the electron temperature in the transient regime reaches some significant values, but that eventually stabilizes to values slightly below 10 eV, which are reasonable for these pressures. The neutral gas temperatures, initialized at 350 K, which is also the wall's temperature, show little variation in the present testcase. In (b), we observe that the dissociation of molecular nitrogen (n_{N_2}) and molecular oxygen (n_{O_2}) is rather important. Both atomic nitrogen and oxygen densities, n_{N} and n_{O} respectively, acquire relative relevance when compared to their

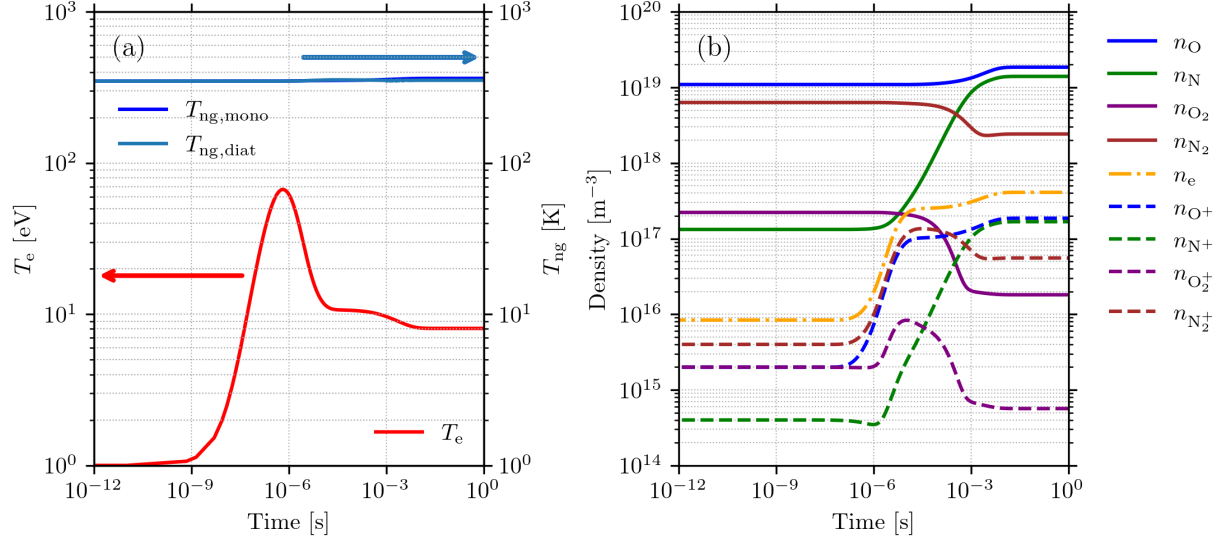


FIG. 4: Temporal evolution of electron and gas temperatures (a) and of plasma species and neutral gas densities (b) at 220 km for an input RF power of 1500 W

Dimensions	Grid settings	Coil properties
$R = 6 \text{ cm}$	$\beta_{\text{ion}} = 0.4, \beta_{\text{neu}} = 0.15$	$N = 5 \text{ turns}$
$L = 10 \text{ cm}$	$V_{\text{acc}} = 1000 \text{ V}$	$f = 13.56 \text{ MHz}$
$A_{\text{intake}} = 1 \text{ m}^2$	$s = 0.1 \text{ cm}$	$R_{\text{coil}} = 2 \Omega$

TABLE III: System parameters, inspired by Refs^{19,20}.

molecular counterparts.

The final compression level, defined as the ratio between the sum of the partial pressures inside and outside the intake, is on the order of 3500. This value is not compatible with a purely passive intake, which would typically require compression levels approximately one order of magnitude lower¹. However, certain active compression designs suggest that the obtained ratio may be achievable^{27,47}.

B. Input RF-power variation

The results presented in this section use the parameters in Table III at an altitude of 220 km. The quantities are presented as a function of the RF power provided by the power supply, which

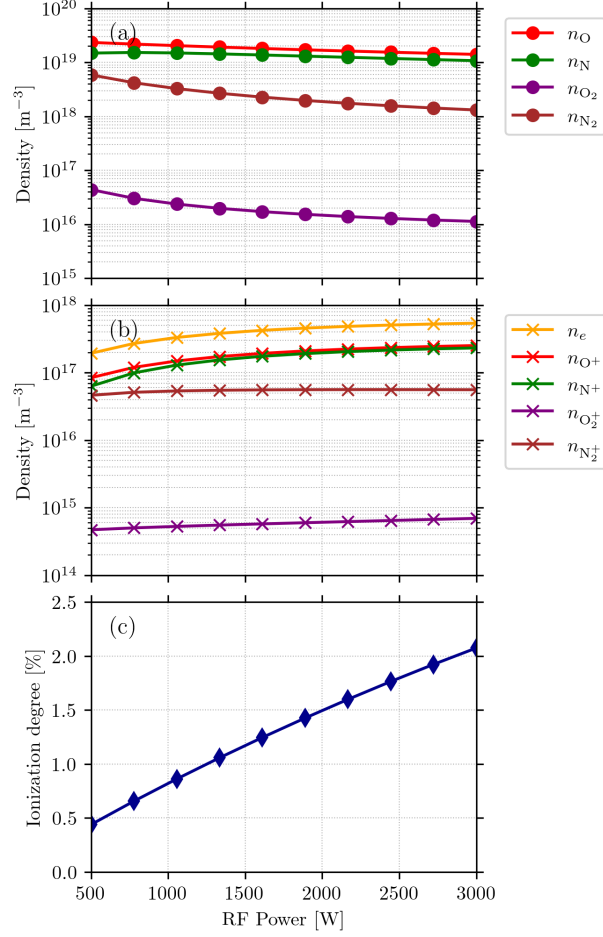


FIG. 5: Variation of the steady-state neutral gas densities (a), and charged species densities (b) at 220 km for different input RF powers. Subfigure (c) shows the ionization degree.

is fixed by the user.

Figure 5 shows the density of atoms and molecules (a) and charged particles (b) in the thruster chamber. Even if the absolute values are lower, the trend is similar to that of xenon. The ionization degree, defined as the ratio of the number of electrons per neutral particle, is shown in (c). This fraction grows as the RF power does, but it remains relatively small (i.e. below 2.0%) in the power range studied.

In general, one should consider the Child-Langmuir limit, which defines the maximal current density that can be extracted between two electrodes. In a single-species plasma, the maximal current reads

$$J_{\text{CL}} = \frac{4}{9} \epsilon_0 \left(\frac{2e}{m_i} \right)^{1/2} \frac{V_{\text{grid}}^{3/2}}{s^2}. \quad (16)$$

The results in xenon¹⁹ and iodine²⁰ for $V_{\text{grid}} = 1000 \text{ V}$ and grids' spacing $s = 0.1 \text{ cm}$ are $J_{\text{CL}}^{\text{Xe}} \sim$

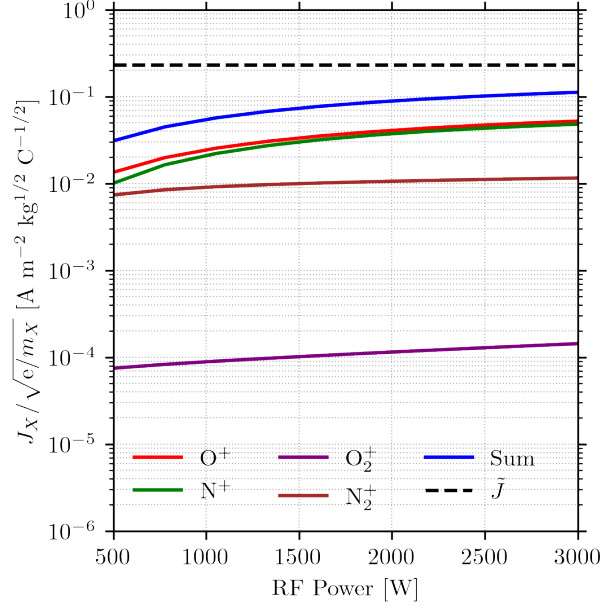


FIG. 6: The current extracted by the grid for each species X divided by $\sqrt{e/m_X}$. $\tilde{J}$ is defined in Eq. (17). Here we consider $V_{\text{grid}} = 1200 \text{ V}$, slightly larger than the plasma-to-neutralizer voltage difference.

150 A/m^2 and $J_{\text{CL}}^{\text{I}} \sim 153 \text{ A/m}^2$, respectively. These values limit the operation of the thruster.

In the case of a multi-species plasma⁴⁸, one can show that the previous equation becomes

$$\sum_X \frac{J_X}{\sqrt{q_X/m_X}} \leq \frac{4}{9} \epsilon_0 \sqrt{2} \frac{V_{\text{grid}}^{3/2}}{s^2} \doteq \tilde{J}, \quad (17)$$

where J_X is the current carried by the species X with a charge q_X . As shown in Figure 6, the lighter masses of nitrogen and oxygen compared to xenon and iodine prevent the Child-Langmuir limit from being reached at any RF power level. The sum of the species' current densities remains at least two times smaller than the limit. Consequently, this does not impose any constraint on the power supplied to the plasma.

When we increase the RF power, the dissociation degree of molecular nitrogen and oxygen increases significantly. This is consistent with the observation of a larger electron temperature, see Figure 7. The plasma density also increases with the power. The increment is significant at low power, while it declines at larger power. In Figure 5 (b), we observe that the increase in plasma density is sustained by atomic ions (i.e. N^+ and O^+), while the molecular ion density (i.e. N_2^+ and O_2^+) varies only slightly with power.

Varying the input RF power by a factor of six results in a decrease of about 40% in atomic

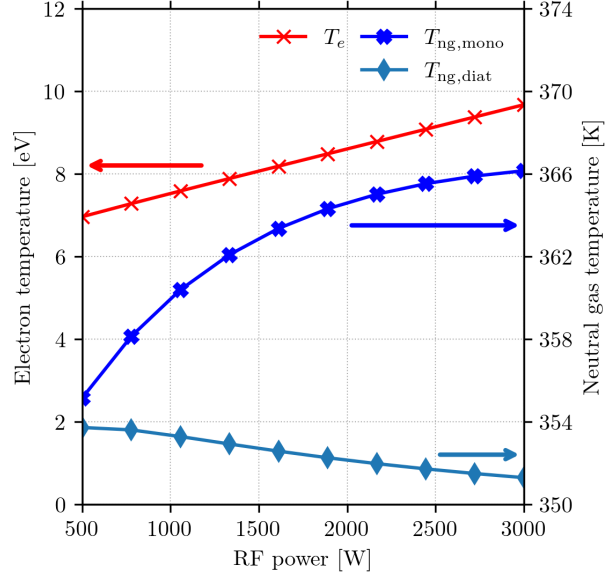


FIG. 7: Electron temperature (red) and neutral gas temperatures (blue) as a function of the RF power.

oxygen density and about 30% in atomic nitrogen density. So, the drop in the neutral density is significantly lower than that observed for xenon¹⁹. This is consistent with the almost constant atomic and molecular species temperature that we observe in Figure 7. The outgoing neutral flux, which mostly depends on the neutral temperature, does not vary particularly, resulting in a small drop of the neutrals inside the thruster.

In the present case, we observe that the heavy species temperatures are close to that of the walls. This is consistent with the higher thermal coefficient of nitrogen and oxygen, which is one order of magnitude larger than that of xenon. From Equation (15), it follows that as κ increases, the gas temperature approaches the wall temperature.

The electron temperature increases when the RF power does. Even if this fact has been observed in several other global models^{19,20}, in this air plasma, the electron temperature reaches values larger than before at around 10 eV. As already suggested in Section IV A, the increase in temperature is related to the lower pressure regime (i.e. around 0.9 mTorr) of the discharge.

Even if the ionization degree is of the order of 1%, as one can see in Figure 5 (c), the thrust produced by the ions is significantly larger than that of neutrals, see Figure 8. When the input RF power increases from 500 W to 3000 W, the thrust increases by a factor of ≈ 3 . However, at all RF powers, the thrust absolute values remain low if compared to those produced by heav-

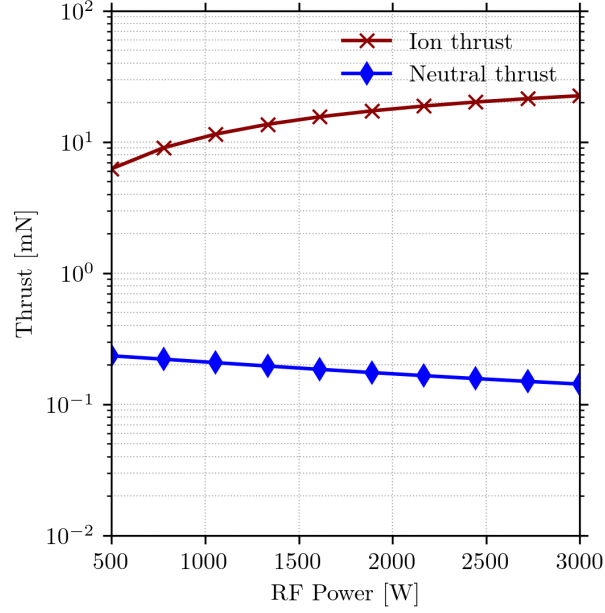


FIG. 8: Thrust produced by ions (red) and by neutrals (blue).

ier atoms^{19,20}. The atmospheric air at 220 km in altitude produces some 20 mN of thrust when operated at 3000 W.

Figure 9 presents some parameters that we use to better understand the discharge characteristics. In (a) we show the thrust power efficiency (i.e. γ), the inductively coupled plasma power transfer efficiency (i.e. ζ), the mass utilization efficiency (i.e. η_a), and the total efficiency (i.e. η_T) defined as follows:

$$\gamma = \frac{P_i + P_N}{P_i + P_{RF}}, \quad (18)$$

$$\zeta = \frac{R_{ind}}{R_{ind} + R_{coil}}, \quad (19)$$

$$\eta_a = \frac{\text{Ion exiting mass flow}}{\text{Neutral entering mass flow}} = \frac{\sum_j m_j \Gamma_j^{\text{ion,out}}}{\sum_j m_j Q_j}, \quad (20)$$

$$\eta_T = \frac{(\sum_X F_{T,X^+})^2}{2(\sum_j m_j Q_j)(P_i + P_{RF})}, \quad (21)$$

In the previous expression, P_i and P_N , are the thrust power generated by ions and neutrals, respectively, defined in Sec III D. $\Gamma_j^{\text{ion,out}}$ is the number of particles of species j ejected for unit time by the grids, creating the thrust. Q_j is the particle injection rate of species j . Although γ and η_T are closely related, they are not equivalent. In contrast to γ , the total thrust efficiency η_T incorporates the effect of the mass-utilization efficiency.

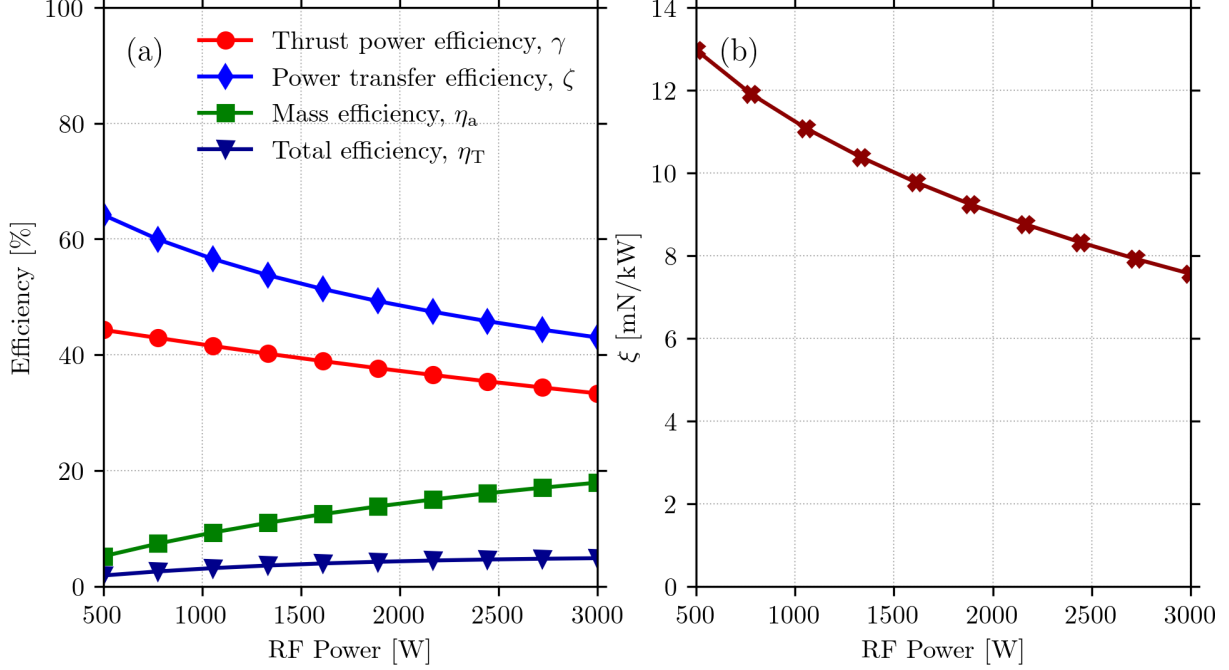


FIG. 9: γ , ζ , η_a , and η_T efficiencies (a) and ξ efficiency (b) as a function of the RF power.

The power transferred to the plasma is not the power delivered by the RF generator because a fraction of the power is dissipated in the coil. The parameter ζ evaluates the efficiency of power coupling. In our case, we observe that ζ has a limited decrease: its value remains slightly above 30%, which is lower than the usual values for this kind of discharges²⁴. This can be understood by the fact that at lower pressure in the sub-mTorr regime, the total momentum transfer collision frequency decreases, worsening the RF power coupling with the plasma.

The thrust power efficiency is the ratio between the thrust power and the total power to maintain the discharge and accelerate the ions. This value, relatively low, tells us that the cost of maintaining the ICP discharge is high when compared to the cost of accelerating the ions outwards. Moreover, the mass efficiency increases with the RF power, but remains below 20%. This last result confirms the low ionization degree reported in Figure 5 (c). This is strictly related to the low η_T , which remains always below 5%.

The quantity ξ , defined as

$$\xi = \frac{F_{T,\text{tot}}}{P_{\text{RF}}},$$

in Figure 9 (b) assesses the thrust delivered per unit of RF power. Since the thrust increment is less than linear with the increment of RF power, ξ decays when the RF power increases, reducing

the thrust generated for a unit of power.

C. Altitude variation

In order to estimate whether a gridded ion thruster can be used in very low orbits in air-breathing mode, in this section we make a simple estimate of the drag force acting on a satellite, and we verify which conditions allow a complete drag compensation. Clearly, the effective drag is strongly dependent on the satellite design. Thus, several models to estimate the drag are possible. The simple model that we have chosen here, inspired by the formulation proposed by Tisaev *et al.*¹⁵, allows us to infer the expected order of magnitude of the drag and its variation with the altitude. The estimation of the drag is simplified, and more refined consideration of the whole ABEP system can be performed¹⁶. However, this is beyond the focus of the current work. Hence, a detailed analysis based on each spacecraft's characteristics should be performed to evaluate the effective drag in a specific mission.

The analysis is conducted under the following assumptions. The satellite operates in a circular orbit around the Earth at an altitude between 100 km and 300 km, which ensures a constant relative velocity between the satellite and atmospheric particles. The exposure to solar radiation and the electrical power generated by the solar panels are assumed to remain constant. The satellite's geometry is cylindrical, with flat solar panels attached to the lengthwise sides of it. The panels are perpendicular to the cylinder's surface and are considered to always be oriented parallel to the relative flow of air, along the main axis of the cylinder that contains the thruster. The flow naturally comes through the inlet, which is one of the bases of the cylinder, and is propelled through the outlet on the other side. Furthermore, the axes of motion, the cylinder's axis, and the direction of the incoming airflow are aligned so that the intake cross-section is perpendicular to the incoming airflow. These assumptions are rather general, however, as already said, they should be adapted when dealing with a specific satellite design.

The total drag is calculated as the sum of the aerodynamic drag acting on the intake, solar arrays, and spacecraft body, together with the momentum loss associated with the deceleration of the collected atmospheric flow.¹⁵ A one-dimensional steady-state equation of motion along the orbital velocity direction is formulated for the spacecraft by imposing force equilibrium between thrust and aerodynamic losses. Under the assumption that the thrust vector is aligned with the

flight direction, the axial force balance gives

$$F_D = \dot{m}_t u_\infty + F_{D,i} + F_{D,s} + F_{D,a}, \quad (22)$$

where F_D is the total drag force, $\dot{m}_t u_\infty$ represents the momentum loss associated with decelerating the collected atmospheric flow, with $\dot{m}_t = \eta_c \dot{m}_\infty$ and $\dot{m}_\infty$ the freestream mass flow rate incident on the intake, from the freestream velocity u_∞ to stagnation conditions within the intake. The quantities $F_{D,i}$, $F_{D,s}$, and $F_{D,a}$ denote the intake pressure drag, spacecraft body drag, and solar-array drag, respectively.

Expressing the drag forces in terms of their aerodynamic coefficients yields

$$F_D = \dot{m}_t u_\infty + \frac{1}{2} \rho_\infty u_\infty^2 [(1 - \eta_c) C_{D,i} A_{\text{intake}} + C_{D,s} A_s + C_{D,a} A_a], \quad (23)$$

where ρ_∞ is the freestream atmospheric density, η_c is the intake collection efficiency, $C_{D,i}$, $C_{D,s}$, and $C_{D,a}$ are the drag coefficients of the intake, spacecraft structure, and solar arrays, respectively, and A_{intake} , A_s , and A_a are the corresponding reference areas. The intake pressure drag originates from the fraction of the incident atmospheric flow that is not captured by the intake. Consequently, the effective frontal area exposed to the freestream is $(1 - \eta_c) A_{\text{intake}}$. We consider a spacecraft aspect ratio $\chi = A_s / 4 A_{\text{intake}}$.

The parameters $C_{D,a}$ and $C_{D,s}$ are fixed at 0.1 and 0.06, respectively, following Tisaev *et al.*¹⁵. The intake drag coefficient is taken as $C_{D,i} = 2.3$, while the spacecraft aspect ratio is assumed to be $\chi = 3$ ¹⁵. The freestream velocity u_∞ is calculated assuming a circular orbit and no atmospheric wind relative to the Earth, using the orbital altitude in Eq. (1). The atmospheric density ρ_∞ is obtained from the NRLMSISE-00 model. Based on the work of Tisaev *et al.*¹⁵, the intake collection efficiency is fixed at $\eta_c = 35\%$. We recall that the intake area is assumed to be 1.0 m^2 . The total solar-array area depends on the spacecraft design and operating conditions and is typically of the order of a few square metres.¹⁵

In Figure 10, we show the variation of the thrust as a function of the altitude for different RF power levels. In the same figure, we report a drag evaluation by Equation (23) at the corresponding altitude conditions. The results are plotted for three different values of A_a : 3 m^2 , 6 m^2 , and 10 m^2 . This area depends on the amount of power required by the satellite and it is specific to every mission. To obtain a rough estimate of the power generated by the solar arrays, a specific power of 368 W, m^{-2} and a total panel area of 4 m^2 mounted on the satellite sides are assumed, following¹⁵. For the three values of A_a considered above, this yields available electrical powers of

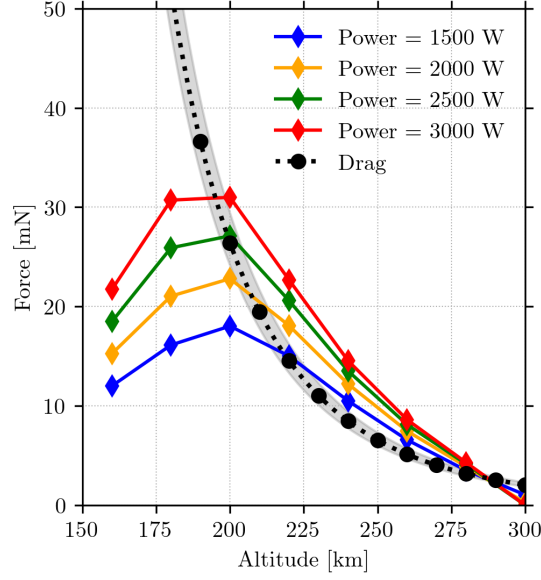


FIG. 10: Thrust, from the 0D model, produced at four different RF power levels and drag, estimated by Equation (23), as a function of the altitude. The dotted line represents the drag for $A_a = 6 \text{ m}^2$, while the shaded area shows the drag variation when A_a is 10 m^2 (higher drag line) and 3 m^2 (lower drag line).

2576 W, 3680 W, and 5152 W, respectively. As one can observe, the drag increases significantly at low altitude. For an altitude higher than approximately 200 km, the thrust produced by the gridded engine is larger than the drag. At an altitude of the order of 300 km, the drag is only a few mN.

The thrust produced by the gridded engine has a maximum between 180 and 200 km, decreasing afterwards. This can be explained by using the efficiencies in Figure 11. At low altitude, the neutral density is relatively high, so the RF power is efficiently transferred to the plasma. However, most of this power is lost in heating the neutral gas. As a consequence, the mass efficiency remains close to zero, making the use of the thruster impractical. The ζ and η_a efficiencies have an opposite behavior when the altitude increases: the first one drops, while the second one increases, remaining, however, rather low if compared to standard fuels. On one hand, this corresponds to a worse energy transfer from the coil to the plasma; on the other hand, it implies that the energy is mostly transferred to the electrons, that increase the ionization. When the atmospheric gas becomes even more rarefied, above 250 km, the discharge enters an extremely low-pressure regime (i. e. few hundred μTorr). In this case, the mass efficiency starts to decay, and the thruster stops working properly.

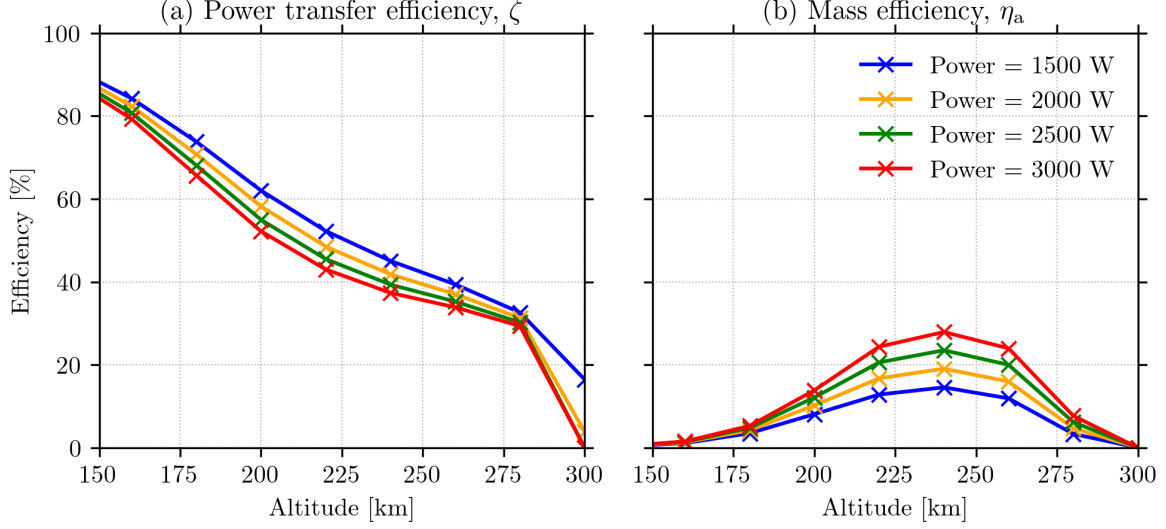


FIG. 11: Power transfer efficiency ζ (a) and mass efficiency η_a (b) as a function of the altitude, for different RF powers.

The study in this section allows us to define a range, going between 200 and 250 km of altitude, in which a gridded ion thruster should balance the drag. Below 200 km, the drag is too large and cannot be compensated. Above 250 km, the thrust exceeds the drag by only a few millinewtons, making satellite operation under these conditions high-risk. Clearly, for any mission design, the drag needs to be estimated precisely in order to assess whether the thruster is effective or not. As said, the calculation here has been done for a fixed atmosphere composition. The atmospheric composition changes depending on latitude and longitude, and over time, and this may have a severe effect on drag and thruster performances. Thus, we recommend considering, for any mission design, the varying atmospheric conditions. We recall that mission design should also account for proper intake design, efficiency variations, available electrical power, and non-circular orbits.

V. CONCLUSIONS

In this work, we presented a global model of a gridded-ion thruster working with atmospheric (nitrogen and oxygen) gas. The model considers particle and energy balance equations for the different species in the plasma. The ions generated by the plasma are accelerated outwards and create the thrust. The simulation results allow us to calculate different metrics of the thruster; in particular, we show how different efficiencies can be calculated. For this study, no particle

backflow toward the intake is assumed. Therefore, active compression is required between the intake and the plasma chamber to maintain a positive pressure gradient and ensure continuous propellant delivery.

The simulation outcomes of our code have been compared to the original paper by Chabert¹⁹, from which our approach is inspired, showing perfect agreement. Later, we use our code to simulate a pure nitrogen discharge, already studied in Ref.¹⁸. Our model successfully reproduces the experimental trends for nitrogen²⁵. Interestingly, we observe that the lower the pressure in the chamber, the higher the electron temperature. This fact is observed both as an experimental result and as a code outcome. A comparison with pure-oxygen discharges²⁵ to validate the model and cross-sections is also performed, showing good agreement.

The analysis of the discharge at fixed altitude (i. e. 220 km) for different RF power levels allows us to show how the plasma behaves in different conditions. Due to the good thermal conductivity of nitrogen and oxygen, the gas temperature is very close to that of the walls. Even if the thrust varies significantly as a function of the RF power, at 3000 W the thrust does not exceed 20 mN. Thrust power efficiency and power transfer efficiencies are both relatively low compared to standard fuels and decrease when the RF power increases. Similarly, the mass efficiency and the ionization degree are both very low, slightly increasing with the power.

The comparison of the thrust and the drag, which is estimated by a simple model, for different altitudes and power levels, shows that ion-gridded thrusters can be used in principle in a range going from around 200 km to 250 km, even if more detailed studies on the specific spacecraft geometry and thruster performances are required. Our analysis shows that at a lower altitude, where the power transfer efficiency is good, the mass efficiency is very low. On the contrary, at around 250 km, the mass efficiency is significantly better, while the power transfer worsens. All in all, we observe that the margin is never huge, but that at orbits of 200 km to 225 km and relatively high power (i. e. 3000 W), the margin could be large enough to fly a satellite.

Although our model provides useful insights about the design of the plasma thruster, it has some intrinsic limitations. First of all, a global model does not resolve the precise spatial distribution of the plasma. Secondly, even if the code has been validated against some nitrogen and oxygen plasma experimental measurements, it would be extremely important to validate it also in atmospheric air conditions. Moreover, for simplicity, we have fixed the atmospheric components. However, the Earth's atmospheric composition depends on latitude and longitude, and on the geomagnetic conditions and solar activity. For a precise mission design, we recommend estimating

those quantities, as well as the drag related to the satellite configuration. Other effects, such as the oxygen corrosion, the spacecraft heating, neutralization, etc., go beyond the scope of the current study, but they should be taken into consideration as well.

The use of a global model does not allow for spatial resolution of plasma-wall interactions, which are critical in this type of discharge. Therefore, more advanced models, such as 2D or 3D simulations as those of Refs^{49–51}, are required to achieve a more accurate description of the plasma behavior. These approaches would enable a more precise evaluation of particle and energy fluxes to the walls, allowing for improved assessment of losses and wall erosion due to ion impact. In particular, the treatment of the temperature jumps at the walls due to thermal accommodation requires integration into the models. Furthermore, accurate modeling of surface processes is necessary to evaluate energy accommodation and ion recombination at the walls, which can significantly influence plasma characteristics and composition. In this context, further experimental work or molecular dynamics simulations may be required to capture these effects with sufficient fidelity.

This study aims to evaluate the potential feasibility of operating a satellite in very low orbit using an ion-gridded thruster. We believe that this study can benefit the whole community in the following way. The code discussed here produces some reasonably good results within seconds: this allows the code's user to easily perform some parametric studies. This study enables an initial assessment of the required size of some configuration components, such as the RF input power, the chamber dimensions, the grid voltage and transparency, etc. This will significantly shorten the thruster's design time.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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